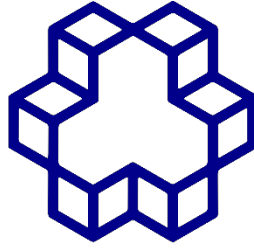


به نام خدا



دانشگاه صنعتی خواجه نصیرالدین طوسی

درس یادگیری ماشین

حل تمرینات سری سوم (SVM)

استاد درس: جناب آقای دکتر ناصرشریف

دانشگاه صنعتی خواجه نصیرالدین طوسی – دانشکده مهندسی کامپیوتر

پاییز ۱۴۰۴

Question 1 (25 marks)

Let the mapping from the input space to a higher dimensional space be

$$x \in R \rightarrow y \equiv \varphi(x) \in R^{2k+1}$$

where

$$\varphi(x) = \left[\frac{1}{\sqrt{2}}, \cos x, \cos 2x, \dots, \cos kx, \sin x, \sin 2x, \dots, \sin kx \right]^T$$

Then show that the corresponding inner product kernel is as following

$$k(x_i, x_j) = y_i^T y_j = \frac{\sin((k+0.5)(x_i - x_j))}{2 \sin\left(\frac{x_i - x_j}{2}\right)}$$

ابتدا يك لم را اثبات مي كنيم كه جدا از ان استفاده خواهد شد:
لم: مر فراضيع ثابت كنيم:

$$\cos(z) + \cos(2z) + \cos(3z) + \dots + \cos(kz) = \frac{\sin((k+0.5)z) - \sin\left(\frac{z}{2}\right)}{2 \sin\left(\frac{z}{2}\right)}$$

$$\sum_{i=1}^k \cos(iz) = \frac{\sin((k+0.5)z) - \sin\left(\frac{z}{2}\right)}{2 \sin\left(\frac{z}{2}\right)} \quad \text{يا به طور معادل:}$$

$$\underbrace{\cos(A)}_{\neq} \underbrace{\sin(B)}_{\neq} = \frac{1}{2} \left[\sin(A+B) - \sin(A-B) \right] \quad \text{اثبات: } \otimes \text{ موداسينم}$$

$$\cos(z) + \cos(2z) + \dots + \cos(kz) = \sum_{i=1}^k \cos(iz)$$

$$= \frac{1}{\sin\left(\frac{z}{2}\right)} \sum_{i=1}^k \cos(iz) \sin\left(\frac{z}{2}\right)$$

$$= \frac{1}{\sin\left(\frac{z}{2}\right)} \sum_{i=1}^k \frac{1}{2} \left[\sin\left(iz + \frac{z}{2}\right) - \sin\left(iz - \frac{z}{2}\right) \right] \quad \text{طبق } \otimes \leftarrow$$

$$= \frac{1}{2 \sin\left(\frac{z}{y}\right)} \sum_{i=1}^K \left[\sin\left(\left(\frac{y_{i+1}}{y}\right)z\right) - \sin\left(\left(\frac{y_{i-1}}{y}\right)z\right) \right]$$

$$= \frac{1}{2 \sin\left(\frac{z}{y}\right)} \left[\left[\cancel{\sin\left(\frac{y}{y}z\right)} - \sin\left(\frac{z}{y}\right) \right] + \left[\cancel{\sin\left(\frac{2y}{y}z\right)} \ominus \cancel{\sin\left(\frac{y}{y}z\right)} \right] + \left[\sin\left(\frac{3y}{y}z\right) \ominus \cancel{\sin\left(\frac{2y}{y}z\right)} \right] \right. \\ \left. + \dots + \left[\cancel{\sin\left(\frac{y_{K-2}}{y}z\right)} - \cancel{\sin\left(\frac{y_{K-1}}{y}z\right)} \right] + \left[\cancel{\sin\left(\frac{y_{K-1}}{y}z\right)} \ominus \cancel{\sin\left(\frac{y_{K-2}}{y}z\right)} \right] + \left[\sin\left(\frac{y_{K+1}}{y}z\right) \ominus \cancel{\sin\left(\frac{y_K}{y}z\right)} \right] \right]$$

$\uparrow$ $i=1$ $\uparrow$ $i=2$ $\uparrow$ $i=3$
 $\uparrow$ $i=K-2$ $\uparrow$ $i=K-1$ $\uparrow$ $i=K$

$$= \frac{1}{2 \sin\left(\frac{z}{y}\right)} \left[-\sin\left(\frac{z}{y}\right) + \sin\left(\frac{y_{K+1}}{y}z\right) \right] = \frac{\sin((K+0.5)\frac{z}{y}) - \sin\left(\frac{z}{y}\right)}{2 \sin\left(\frac{z}{y}\right)}$$

اگر رقم کسیر صفری می شود حد چپ + با چپ - سه نام جلوتر (یعنی برشته بعدی!) خط میفرود و حذف می شود.
 بنابراین تنها جملات حایلیت شده باقی می ماند.

$$\Rightarrow \sum_{i=1}^K \cos(iz) = \frac{\sin((K+0.5)z) - \sin\left(\frac{z}{2}\right)}{2 \sin\left(\frac{z}{2}\right)} \quad \blacksquare$$

حال فعلی راستی این لم به اثبات سوال می پردازیم :

$$\left. \begin{aligned} \cos(A)\cos(B) &= \frac{1}{2} [\cos(A+B) + \cos(A-B)] \\ \sin(A)\sin(B) &= \frac{1}{2} [\cos(A-B) - \cos(A+B)] \end{aligned} \right\} \text{ یادآور ① می دانیم :}$$

Question 1 (25 marks)

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$$x \in R \rightarrow y \equiv \varphi(x) \in R^{2k+1}$$

where

$$\varphi(x) = \left[\frac{1}{\sqrt{2}}, \cos x, \cos 2x, \dots, \cos kx, \sin x, \sin 2x, \dots, \sin kx \right]^T$$

Then show that the corresponding inner product kernel is as following

$$k(x_i, x_j) = y_i^T y_j = \frac{\sin((k+0.5)(x_i - x_j))}{2 \sin(\frac{x_i - x_j}{2})}$$

Proof:

$$K(x_i, x_j) = y_i^T y_j = \varphi(x_i)^T \varphi(x_j) =$$

$$= \left[\frac{1}{\sqrt{2}}, \cos(x_i), \cos(2x_i), \dots, \cos(kx_i), \sin(x_i), \sin(2x_i), \dots, \sin(kx_i) \right]$$

$$\begin{bmatrix} \frac{1}{\sqrt{2}} \\ \cos(x_j) \\ \cos(2x_j) \\ \vdots \\ \cos(kx_j) \\ \sin(x_j) \\ \sin(2x_j) \\ \vdots \\ \sin(kx_j) \end{bmatrix}$$

$$= \left(\frac{1}{\sqrt{2}} \right)^2 + \cos(x_i) \cos(x_j) + \cos(2x_i) \cos(2x_j) + \dots + \cos(kx_i) \cos(kx_j) \\ + \sin(x_i) \sin(x_j) + \sin(2x_i) \sin(2x_j) + \dots + \sin(kx_i) \sin(kx_j)$$

$$= \frac{1}{2} + \sum_{n=1}^K \cos(n x_i) \cos(n x_j) + \sum_{n=1}^K \sin(n x_i) \sin(n x_j) \quad \leftarrow \text{طبق بايادورس 1}$$

$$= \frac{1}{2} + \sum_{n=1}^K \frac{1}{2} \left[\cos(n x_i + n x_j) + \cos(n x_i - n x_j) \right] + \sum_{n=1}^K \frac{1}{2} \left[\cos(n x_i - n x_j) - \cos(n x_i + n x_j) \right]$$

$$= \frac{1}{2} + \frac{1}{2} \sum_{n=1}^K \left[\cancel{\cos(n(x_i + x_j))} + \cos(n(x_i - x_j)) + \cos(n(x_i - x_j)) - \cancel{\cos(n(x_i + x_j))} \right]$$

$$= \frac{1}{2} + \frac{1}{2} \sum_{n=1}^K \left[2 \cos(n(x_i - x_j)) \right]$$

$$= \frac{1}{\gamma} + \sum_{n=1}^K \cos(nz) \quad , \quad z = x_i - x_j$$

$$= \frac{1}{\gamma} + \frac{\sin((K+0, \omega)z) - \sin(\frac{z}{\gamma})}{\gamma \sin(\frac{z}{\gamma})}$$

← طبق لم قبل

$$= \frac{1}{\gamma} \left[\frac{\cancel{\sin(\frac{z}{\gamma})} + \sin((K+0, \omega)z) - \cancel{\sin(\frac{z}{\gamma})}}{\sin(\frac{z}{\gamma})} \right]$$

$$= \frac{\sin((K+0, \omega)z)}{\gamma \sin(\frac{z}{\gamma})} = \frac{\sin[(K+0, \omega)(x_i - x_j)]}{\gamma \sin(\frac{x_i - x_j}{\gamma})}$$



Question 2 (25 marks)

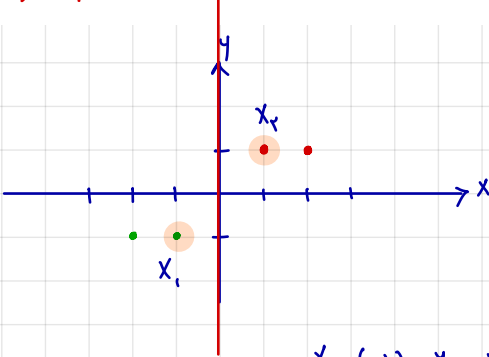
Two different one-dimensional classification tasks are given in the following tables. Draw the two-class data points in an (x, y) plane. (Draw two separate graphs.) Find analytically and sketch the optimal canonical hyperplanes belonging to these two classification tasks. Determine the equations for decision boundaries. (Hint: Identify the support vector first) (25 Mark)

a. x	y = d
• 2	1
• -1	-1
• -2	-1
• 1	1

b. x	y = d
3	1
1	-1
-1	-1

a-2

معادله خط جدا ساز $x=0$



نکته: ترمیم داده ها باید داده ها یک به یک هستند!

واقع است که Support vector ها نقاط مرز هستند

که در شکل ها علامت شده اند

Support vectors

$$\begin{aligned} X_i = (-1), y_i = -1 & \xrightarrow{\text{افزایش کردن تا 1}} X_i = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, y_i = -1 \\ X_v = (+1), y_v = +1 & \xrightarrow{\text{باینری}} X_v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, y_v = +1 \end{aligned}$$

$$K_{ij} = X_i^T X_j$$

$$K_{ii} = X_i^T X_i = \begin{bmatrix} 1 \\ -1 \end{bmatrix}^T \begin{bmatrix} 1 \\ -1 \end{bmatrix} = (1)(1) + (-1)(-1) = 1 + 1 = 2 \Rightarrow K_{ii} = 2$$

$$K_{vv} = X_v^T X_v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}^T \begin{bmatrix} 1 \\ 1 \end{bmatrix} = (1)(1) + (1)(1) = 1 + 1 = 2 \Rightarrow K_{vv} = 2$$

$$K_{vi} = K_{iv} = X_i^T X_v = \begin{bmatrix} 1 \\ -1 \end{bmatrix}^T \begin{bmatrix} 1 \\ 1 \end{bmatrix} = (1)(1) + (-1)(1) = 1 - 1 = 0 \Rightarrow K_{vi} = K_{iv} = 0$$

$$L_D(\alpha) = \sum_{i=1}^Y \alpha_i - \frac{1}{Y} \sum_{i=1}^Y \sum_{j=1}^Y \alpha_i \alpha_j y_i y_j K_{ij}$$

$$= \alpha_i + \alpha_v - \frac{1}{Y} \left[\underbrace{\alpha_i^T y_i^T}_{(-1)^2} K_{ii} + \underbrace{\alpha_v^T y_v^T}_{(1)^2} K_{vv} + \cancel{2 \alpha_i \alpha_v y_i y_v K_{iv}} \right]$$

$K_{iv} = 0$

$$= \alpha_1 + \alpha_y - \frac{1}{y} \left[\alpha_1^y (-1)^y (y) + \alpha_y^y (1)^y (y) + 0 \right]$$

$$= \alpha_1 + \alpha_y - \frac{1}{y} \left[y \alpha_1^y + y \alpha_y^y \right] = \alpha_1 + \alpha_y - (\alpha_1^y + \alpha_y^y)$$

$$\Rightarrow L_D(\alpha) = \alpha_1 + \alpha_y - (\alpha_1^y + \alpha_y^y)$$

$$\frac{\partial L_D(\alpha)}{\partial \alpha_1} = 0 \Rightarrow 1 - y \alpha_1 = 0 \Rightarrow \boxed{\alpha_1 = \frac{1}{y}}$$

$$\frac{\partial L_D(\alpha)}{\partial \alpha_y} = 0 \Rightarrow 1 - y \alpha_y = 0 \Rightarrow \boxed{\alpha_y = \frac{1}{y}}$$

$$w = \sum_{i=1}^y \alpha_i y_i \vec{x}_i = \alpha_1 y_1 \vec{x}_1 + \alpha_y y_y \vec{x}_y = \left(\frac{1}{y}\right)(-1) \begin{bmatrix} 1 \\ -1 \end{bmatrix} + \left(\frac{1}{y}\right)(1) \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \frac{1}{y} \begin{bmatrix} -1 + 1 \\ +1 + 1 \end{bmatrix} = \frac{1}{y} \begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\Rightarrow w = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{array}{l} \leftarrow \text{متوسط } y \text{ (bias)} \\ \leftarrow \text{متوسط } x \end{array}$$

معادله خط جدا ساز $w^T x = 0 \Rightarrow \begin{bmatrix} 0 \\ 1 \end{bmatrix}^T \begin{bmatrix} 1 \\ x \end{bmatrix} = 0 \Rightarrow 0 + x = 0 \Rightarrow \boxed{x = 0}$

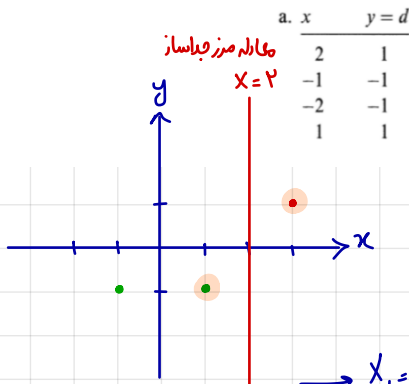
Decision Rule: $y(x) = \text{Sign}(w^T x) = \text{Sign}(x) \rightarrow$ تعیین کلاس هر نمونه جدید بر اساس این رابطه

$$\left\{ \begin{array}{l} x=1 \rightarrow y(1) = \text{Sign}(1) = 1 \checkmark \\ x=-1 \rightarrow y(-1) = \text{Sign}(-1) = -1 \checkmark \end{array} \right\}$$

ضرورت فرگیزد. مثلاً $\leftarrow$

Question 2 (25 marks)

Two different one-dimensional classification tasks are given in the following tables. Draw the two-class data points in an (x, y) plane. (Draw two separate graphs.) Find analytically and sketch the optimal canonical hyperplanes belonging to these two classification tasks. Determine the equations for decision boundaries. (Hint: Identify the support vector first) (25 Mark)



b. x $y = d$

3	1
1	-1
-1	-1

b-2

Support vectors

$$\begin{aligned} X_1 &= (2) , y_1 = +1 \xrightarrow{\text{bias افزودن}} X_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} , y_1 = +1 \\ X_2 &= (1) , y_2 = -1 \xrightarrow{\sim \sim} X_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} , y_2 = -1 \end{aligned}$$

$$K_{ij} = X_i^T X_j$$

$$K_{11} = X_1^T X_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}^T \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 1 \times 1 + 2 \times 2 = 1 + 4 = 5 \Rightarrow K_{11} = 5$$

$$K_{22} = X_2^T X_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}^T \begin{bmatrix} 1 \\ 1 \end{bmatrix} = (1)(1) + (1)(1) = 1 + 1 = 2 \Rightarrow K_{22} = 2$$

$$K_{12} = K_{21} = X_1^T X_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}^T \begin{bmatrix} 1 \\ 1 \end{bmatrix} = (1)(1) + (2)(1) = 1 + 2 = 3 \Rightarrow K_{12} = 3$$

$$L_D(\alpha) = \sum_{i=1}^2 \alpha_i - \frac{1}{2} \sum_{i=1}^2 \sum_{j=1}^2 \alpha_i \alpha_j y_i y_j K_{ij}$$

$$= \alpha_1 + \alpha_2 - \frac{1}{2} \left[\alpha_1^2 y_1^2 K_{11} + \alpha_2^2 y_2^2 K_{22} + 2 \alpha_1 \alpha_2 y_1 y_2 K_{12} \right]$$

$$= \alpha_1 + \alpha_2 - \frac{1}{2} \left[\alpha_1^2 (1)^2 (5) + \alpha_2^2 (-1)^2 (2) + 2 \alpha_1 \alpha_2 (-1)(1)(3) \right]$$

$$= \alpha_1 + \alpha_y - \frac{1}{Y} \left[10\alpha_1^Y + Y\alpha_y^Y - 11\alpha_1\alpha_y \right]$$

$$\Rightarrow L_D(\alpha) = \alpha_1 + \alpha_y - \left[10\alpha_1^Y + \alpha_y^Y - \sum \alpha_1\alpha_y \right]$$

$$\frac{\partial L_D(\alpha)}{\partial \alpha_1} = 0 \Rightarrow 1 - [10\alpha_1 - \sum \alpha_y] = 0 \Rightarrow 10\alpha_1 - \sum \alpha_y = 1$$

$$\frac{\partial L_D(\alpha)}{\partial \alpha_y} = 0 \Rightarrow 1 - [Y\alpha_y - \sum \alpha_1] = 0 \Rightarrow -\sum \alpha_1 + Y\alpha_y = 1$$

$$\begin{cases} 10\alpha_1 - \sum \alpha_y = 1 \\ -\sum \alpha_1 + Y\alpha_y = Y \end{cases} \xrightarrow{xy} \Rightarrow Y\alpha_1 = Y \Rightarrow \boxed{\alpha_1 = \frac{Y}{Y}}$$

$$-\sum \alpha_1 + Y\alpha_y = 1 \xrightarrow{\alpha_1 = \frac{Y}{Y}} -\sum \left(\frac{Y}{Y}\right) + Y\alpha_y = 1 \Rightarrow Y\alpha_y = Y+1 \Rightarrow \boxed{\alpha_y = \frac{Y}{Y}}$$

$$W = \sum_{i=1}^Y \alpha_i y_i \vec{X}_i = \alpha_1 y_1 \vec{X}_1 + \alpha_y y_y \vec{X}_y = \frac{Y}{Y} (+1) \begin{bmatrix} 1 \\ Y \end{bmatrix} + \frac{Y}{Y} (-1) \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \frac{1}{Y} \begin{bmatrix} Y-1 \\ Y-1 \end{bmatrix} = \frac{1}{Y} \begin{bmatrix} -1 \\ Y \end{bmatrix} = \begin{bmatrix} -Y \\ 1 \end{bmatrix}$$

$$\Rightarrow W = \begin{bmatrix} -Y \\ 1 \end{bmatrix} \begin{matrix} \leftarrow \text{مستطربا bias} \\ \leftarrow \text{مستطربا } x \end{matrix}$$

$$\text{معادله خط کلاسز : } W^T X = 0 \Rightarrow \begin{bmatrix} -Y \\ 1 \end{bmatrix}^T \begin{bmatrix} 1 \\ x \end{bmatrix} = 0 \Rightarrow -Y + x = 0 \Rightarrow \boxed{x = Y}$$

$$\text{Decision Rule: } y(x) = \text{Sign}(W^T X) = \text{Sign}(-Y + x)$$

کلاسز کلاسز

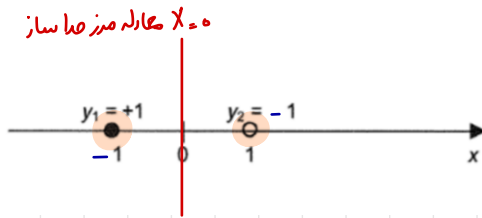
$$x = Y \rightarrow y(Y) = \text{Sign}(-Y + Y) = \text{Sign}(0) = +1 \checkmark$$

$$x = +1 \rightarrow y(1) = \text{Sign}(-Y + 1) = \text{Sign}(-1) = -1 \checkmark$$

$$x = -1 \rightarrow y(-1) = \text{Sign}(-Y - 1) = \text{Sign}(-2) = -1 \checkmark$$

Question 3 (25 marks)

Two one-dimensional data shown in the following figure should be classified using SVM with the first-order polynomial kernel. Compute Kernel Matrix and Find α 's and decision boundary using SVM Equations.



SVM با کرنل چندجمله ای درجه P → P -order Polynomial Kernel

$$K_{ij} = (X_i^T X_j + 1)^P$$

$$P=1 \Rightarrow K_{ij} = X_i^T X_j + 1$$

چون از هر کلاس فقط یک نقطه داریم عدد Support vector.

$$X_1 = (-1), y_1 = +1 \xrightarrow{\text{افزایش کردن (bias)}} X_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, y_1 = +1$$

$$X_2 = (+1), y_2 = -1 \xrightarrow{\sim \sim \sim} X_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, y_2 = -1$$

$$K_{ij} = X_i^T X_j + 1$$

محاسبه ماتریس Kernel :

$$K_{11} = X_1^T X_1 + 1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}^T \begin{bmatrix} 1 \\ -1 \end{bmatrix} + 1 = (1+1) + 1 = 3$$

$$K_{22} = X_2^T X_2 + 1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}^T \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 1 = (1+1) + 1 = 3$$

$$K_{12} = K_{21} = X_1^T X_2 + 1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}^T \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 1 = \overbrace{(1-1)}^0 + 1 = 1$$

$$\Rightarrow K = \begin{bmatrix} K_{11} & K_{1r} \\ K_{r1} & K_{rr} \end{bmatrix} = \begin{bmatrix} w & 1 \\ 1 & w \end{bmatrix}$$

$$L_D(\alpha) = \sum_{i=1}^r \alpha_i - \frac{1}{r} \sum_{i=1}^r \sum_{j=1}^r \alpha_i \alpha_j y_i y_j K_{ij}$$

$$= \alpha_1 + \alpha_r - \frac{1}{r} \left[\alpha_1 y_1^r K_{11} + \alpha_r y_r^r K_{rr} + r \alpha_1 \alpha_r y_1 y_r K_{1r} \right]$$

$$= \alpha_1 + \alpha_r - \frac{1}{r} \left[w \alpha_1 + w \alpha_r - r \alpha_1 \alpha_r \right]$$

$$\left. \begin{aligned} \frac{\partial L}{\partial \alpha_1} = 0 &\Rightarrow 1 - \frac{1}{r} [r \alpha_1 - r \alpha_r] = 0 \Rightarrow w \alpha_1 - \alpha_r = 1 \\ \frac{\partial L}{\partial \alpha_r} = 0 &\Rightarrow 1 - \frac{1}{r} [r \alpha_r - r \alpha_1] = 0 \Rightarrow w \alpha_r - \alpha_1 = 1 \end{aligned} \right\}$$

$$\Rightarrow \begin{cases} w \alpha_1 - \alpha_r = 1 \\ -w \alpha_1 + r \alpha_r = w \end{cases} \xrightarrow{+} \wedge \alpha_r = \varepsilon \Rightarrow \boxed{\alpha_r = \frac{1}{r}}$$

$$w \alpha_r - \alpha_1 = 1 \Rightarrow \alpha_1 = \frac{w}{r} - 1 = \frac{1}{r} \Rightarrow \boxed{\alpha_1 = \frac{1}{r}}$$

$$\Rightarrow \boxed{\alpha_1 = \alpha_r = \frac{1}{r}}$$

$$w = \sum_{i=1}^2 \alpha_i y_i \vec{x}_i = \alpha_1 y_1 \vec{x}_1 + \alpha_2 y_2 \vec{x}_2 = \frac{1}{2} (1) \begin{bmatrix} 1 \\ -1 \end{bmatrix} + \frac{1}{2} (-1) \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \frac{1}{2} \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 0 \\ -2 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

$$\Rightarrow w = \begin{bmatrix} 0 \\ -1 \end{bmatrix} \begin{array}{l} \leftarrow \text{مستطرد با bias} \\ \leftarrow \text{مستطرد با } x \end{array}$$

معادله خط جدا ساز: $w^T x = 0 \Rightarrow \begin{bmatrix} 0 \\ -1 \end{bmatrix}^T \begin{bmatrix} 1 \\ x \end{bmatrix} = \underbrace{0 - x = 0}_{\text{مستطرد با } x} \Rightarrow x = 0$

Decision Rule: $y(x) = \text{Sign}(w^T x) = \text{Sign}(-x)$

$$\begin{cases} x = -1 \rightarrow y(-1) = \text{Sign}(-(-1)) = \text{Sign}(+1) = +1 \checkmark \\ x = +1 \rightarrow y(+1) = \text{Sign}(-1) = -1 \checkmark \end{cases}$$

Question 4 (25 marks)

Using the KKT stationary conditions, find the minimum of the function f subject to the following constraints. Check your answer graphically

$$f(x_1, x_2) = (x_1 - 1)^2 + (x_2 - 2)^2$$

$$x_2 - x_1 = 1$$

$$x_2 + x_1 \leq 2$$

$$x_2 \geq 0 \quad x_1 \geq 0$$

اداموس مقینه KKT

(Kuhn-Tucker Theorem)

Given an optimization problem with convex domain $\Omega \subseteq \mathbb{R}^N$

$$\begin{array}{ll} \text{Minimize} & f(z) \quad z \in \Omega \\ \text{Subject to} & g_i(z) \leq 0 \quad i = 1..k \\ & h_i(z) = 0 \quad i = 1..m \end{array}$$

- with $f \in C^1$ convex and g_i, h_i affine, necessary & sufficient conditions for a normal point z^* to be an optimum are the existence of α^*, β^* such that

$$\begin{array}{l} (*) \left\{ \begin{array}{l} \partial L(z^*, \alpha^*, \beta^*) / \partial z = 0 \\ \partial L(z^*, \alpha^*, \beta^*) / \partial \beta = 0 \\ \alpha_i^* g_i(z^*) = 0 \quad i = 1..k \\ g_i(z^*) \leq 0 \quad i = 1..k \\ \alpha_i^* \geq 0 \quad i = 1..k \end{array} \right. \quad \text{where:} \quad L(z, \alpha, \beta) = f(z) + \sum_{i=1}^k \alpha_i g_i(z) + \sum_{i=1}^m \beta_i h_i(z) \end{array}$$

- $L(z, \alpha, \beta)$ is known as a *generalized Lagrangian function*
- The third condition is known as the Karush-Kuhn-Tucker (KKT) complementary condition
 - It implies that for active constraints $\alpha_i \geq 0$; and for inactive constraints $\alpha_i = 0$
 - As we will see in a minute, the KKT condition allows us to identify the training examples that define the largest margin hyperplane. These examples will be known as Support Vectors

[Cristianini and Shawe-Taylor, 2000]

برای یافتن ضمیمه تابع f تحت شرایط داده شده ابتدا $L(z, \alpha, \beta)$ را تعریف می‌کنیم. طبق مقینه KKT آن $z = (x_1, x_2)$ است که L ضمیمه می‌کند باید در شروط $(*)$ صدق کند. α, β ضرایب لاگرانژ هستند و مجهول‌اند

$$f(x_1, x_2) = (x_1 - 1)^2 + (x_2 - 2)^2$$

$$x_2 - x_1 = 1 \longrightarrow h(x_1, x_2) = x_2 - x_1 - 1 = 0$$

$$\begin{aligned} x_2 + x_1 &\leq 2 \\ x_2 &\geq 0 \\ x_1 &\geq 0 \end{aligned} \longrightarrow g(x_1, x_2) = x_2 + x_1 - 2 \leq 0$$

شکل دادن L :

$$\begin{aligned} L(x_1, x_2, \alpha, \beta) &= f(x_1, x_2) + \alpha g(x_1, x_2) + \beta h(x_1, x_2) \\ &= (x_1 - 1)^2 + (x_2 - 2)^2 + \alpha(x_2 + x_1 - 2) + \beta(x_2 - x_1 - 1) \end{aligned}$$

شرط KKT :

$$\textcircled{1} \frac{\partial L}{\partial z} = 0 \begin{cases} \frac{\partial L}{\partial x_1} = 0 \Rightarrow 2(x_1 - 1) + \alpha(1) + \beta(-1) = 0 \longrightarrow 2x_1 + \alpha - \beta = 2 \\ \frac{\partial L}{\partial x_2} = 0 \Rightarrow 2(x_2 - 2) + \alpha(1) + \beta(1) = 0 \longrightarrow 2x_2 + \alpha + \beta = 4 \end{cases}$$

$$\textcircled{2} \frac{\partial L}{\partial \beta} = 0 \Rightarrow x_2 - x_1 - 1 = 0$$

$$\textcircled{3} \alpha g(x_1, x_2) = 0 \Rightarrow \alpha(x_2 + x_1 - 2) = 0$$

$$\textcircled{4} g(x_1, x_2) \leq 0 \Rightarrow x_2 + x_1 - 2 \leq 0$$

$$\textcircled{5} \alpha \geq 0$$

مسئله را می توان نوشت :

$\Rightarrow$

$$\begin{aligned} \textcircled{1-1} & 2x_1 + \alpha - \beta = 2 \\ \textcircled{1-2} & 2x_2 + \alpha + \beta = 4 \\ \textcircled{2} & x_2 - x_1 - 1 = 0 \\ \textcircled{3} & \alpha(x_2 + x_1 - 2) = 0 \\ \textcircled{4} & x_2 + x_1 - 2 \leq 0 \\ \textcircled{5} & \alpha \geq 0 \end{aligned}$$

با در نظر گرفتن شرط ⑤ یعنی $\alpha \geq 0$ دو حالت داریم :

حالت اول : $\alpha = 0$ باشد . داریم :

$$\left. \begin{array}{l} \text{①-①} \quad 2x_1 + \alpha - \beta = 2 \rightarrow 2x_1 - \beta = 2 \\ \text{①-②} \quad 2x_2 + \alpha + \beta = 2 \rightarrow 2x_2 + \beta = 2 \end{array} \right\} \rightarrow 2(x_1 + x_2) = 4 \rightarrow x_1 + x_2 = 2 \quad (*)$$

$$\text{②} \quad x_2 - x_1 - 1 = 0 \rightarrow x_2 = x_1 + 1 \quad (*)$$

$$\checkmark \text{③} \quad \alpha(x_2 + x_1 - 2) = 0$$

$$\times \text{④} \quad x_2 + x_1 - 2 \leq 0$$

$$\checkmark \text{⑤} \quad \alpha \geq 0$$

$$(*) \quad \left\{ \begin{array}{l} x_1 + x_2 = 2 \rightarrow x_1 + x_1 + 1 = 2 \Rightarrow 2x_1 = 1 \Rightarrow x_1 = \frac{1}{2} \\ x_2 = x_1 + 1 \end{array} \right. \quad x_2 = \frac{3}{2}$$

تفاوتی با شرط ④ $\Rightarrow x_2 + x_1 - 2 = \frac{3}{2} + \frac{1}{2} - 2 = 0 \Rightarrow$ ~~نیست~~

بنابراین فرض $\alpha = 0$ غلط بوده و $\alpha > 0$ است .

حالت دوم : $\alpha > 0$: داریم :

$$\left. \begin{array}{l} \text{③} \quad \alpha(x_2 + x_1 - 2) = 0 \xrightarrow[\alpha \neq 0]{\alpha > 0} x_2 + x_1 - 2 = 0 \\ \text{②} \quad x_2 - x_1 - 1 = 0 \end{array} \right\} \xrightarrow{+} 2x_2 - 3 = 0 \Rightarrow \boxed{x_2 = \frac{3}{2}}$$

$$\text{②} \quad x_2 - x_1 - 1 = 0 \xrightarrow{x_2 = \frac{3}{2}} \frac{3}{2} - x_1 - 1 = 0 \Rightarrow x_1 = \frac{3}{2} - 1 = \frac{1}{2}$$

$$\Rightarrow \boxed{x_1 = \frac{1}{2}}$$

$$\left. \begin{array}{l} \text{①-①} \quad 1 + \alpha - \beta = 2 \\ \text{①-②} \quad 3 + \alpha + \beta = 2 \end{array} \right\} \xrightarrow{+} 4 + 2\alpha = 4 \Rightarrow 2\alpha = 0 \Rightarrow \boxed{\alpha = 0}$$

$$1 + \alpha - \beta = 2 \xrightarrow{\alpha = 0} 1 + 1 - \beta = 2 \Rightarrow \boxed{\beta = 0}$$

$$\Rightarrow \alpha = 1, \beta = 0, \boxed{x_1 = \frac{1}{4}, x_2 = \frac{3}{4}}$$

نیازین تابع $f(x_1, x_2)$ با در نظر گرفتن محدودیت‌های مرز در نقطه $(x_1, x_2) = (\frac{1}{4}, \frac{3}{4})$ مینیمم می‌شود.

برای جواب به دایره رسم شکل:

